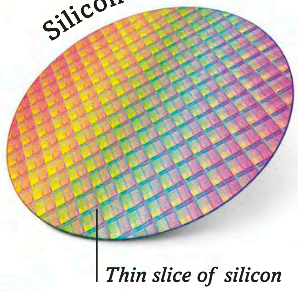
Uses

Opal ring

Opal is a jewel *containing both silicon oxide and water molecules.*

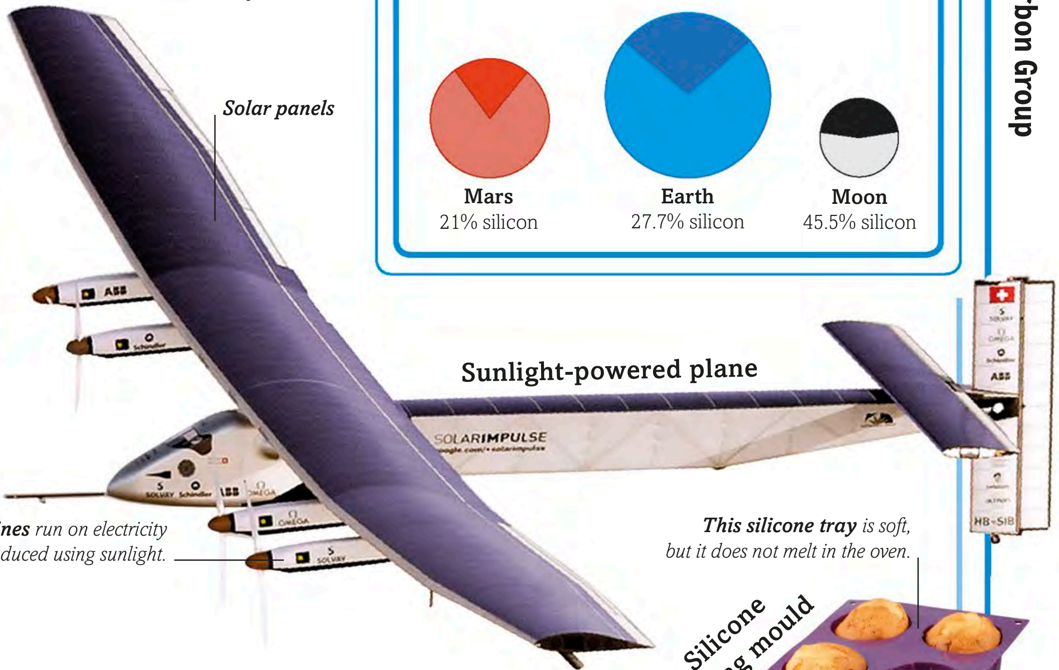


Silicon wafer

Thin slice of silicon

Screens of silicon-based smoke *can be used in battles.*

Smokescreen



Sunlight-powered plane

Solar panels

The engines *run on electricity produced using sunlight.*

SILICON IN SPACE

Earth and Mars contain similar amounts of silicon. This element forms on the outer layer, or crust, of both planets. In contrast, the Moon is almost half silicon. Astronomers think this tells us that the Moon was formed from Earth's surface after an asteroid smashed into our planet about 4.4 billion years ago.

Mars	Earth	Moon
21% silicon	27.7% silicon	45.5% silicon

Ceramic vase

Ceramic *is made of a silicon-based clay that has been shaped and permanently hardened by heat.*

Silicate aerogel in an experiment

Aerogel *conducts the heat from the flame poorly, preventing it from passing to the flower.*

This silicone tray *is soft, but it does not melt in the oven.*

Silicone baking mould

Silicone watch

This silicone band *is flexible and strong.*

One of the most important uses of silicon is in electronics. Thin slices called **silicon wafers** drive electronic circuits. This versatile element is also used to turn sunlight into electricity in solar panels. Artificial silica is used to create **aerogel**, a lightweight but tough substance that does not conduct heat well. It is used in fire-fighting suits, and prevents flames reaching a firefighter. Another silicon compound is silicone, which can be moulded into any shape, and is used in a wide range of products from **baking moulds** to **watches**.